ENN581 Robot Motion, Control and Planning · Mock exam 2 — questions

Indicative marks in [brackets] (the real paper's marks are not published). 2 hours. Calculator and printed notes allowed. Write all working.

Mock exam 2 indicative total 64 marks · 2 hours

Q1 · DH table of an RRP arm derive calculate 6 marks

A base joint rotates about the vertical z_0. At height H = 0.5 m, a second revolute joint with a horizontal axis tilts a boom away from the vertical by θ_2. The boom points straight *up* at $\theta_2 = 0$, and positive θ_2 tilts it outwards, in the direction set by θ_1. Both joint axes meet at one point. A prismatic joint extends the tool along the boom by d_3. (a) Give a proximal DH table [3]. (b) Find the tool position at $\theta_1 = 40°$, $\theta_2 = 30°$, $d_3 = 0.6$ m [3].

Q2 · Inverse kinematics of the RRP arm derive calculate 6 marks

For the arm of Q1, derive the IK and evaluate it for p = (0.3, 0.4, 0.9) m. How many solutions are there?

Q3 · Four-bar: assembly modes calculate sketch explain 8 marks

A four-bar has ground $L_0 = 0.8$ m, input crank $L_1 = 0.3$ m (angle θ_1 at $\{S\} = (0, 0)$), coupler $L_2 = 0.7$ m, rocker $L_3 = 0.5$ m pivoted at $(0.8, 0)$. (a) Find both positions of the coupler–rocker joint B at $\theta_1 = 60°$ and the coupler angle ϕ for each [5]. (b) Can the crank rotate fully? Justify [2]. (c) What is the mobility? [1]

Q4 · Linear segment with parabolic blends `calculate` `sketch` `6 marks`

A joint moves from 10° to 70° in 3 s using LSPB with blend acceleration 30 °/s². (a) Check that the acceleration is large enough [1]. (b) Find the blend time, the linear-segment speed, and the angle at the end of the first blend [3]. (c) Sketch θ, $\dot{\boldsymbol{\theta}}$, $\ddot{\boldsymbol{\theta}}$ against time [2].

Q5 · Jacobian of the W13 RRR arm `derive` `explain` `7 marks`

For the RRR arm of Sample Q1 (position $x = c_1 r$, $y = s_1 r$, $z = L_1 + L_2 + L_3 s_2 + L_4 s_{23}$, with $r = L_3 c_2 + L_4 c_{23}$): (a) derive the 3×3 position Jacobian [3]; (b) show that $\det J = -L_3 L_4 \sin\theta_3 \cdot r$ and interpret each factor [3]; (c) evaluate $\det J$ for $L_3 = 0.4$, $L_4 = 0.3$ m at $\theta = (30°, 20°, 40°)$ [1].

Q6 · Two-link arm with uniform rods derive calculate 8 marks

A planar 2R arm in a vertical plane (q_1 measured from the horizontal, q_2 relative to link 1, gravity along −y) has uniform rods: m_1 = 3 kg, l_1 = 0.6 m; m_2 = 2 kg, l_2 = 0.4 m (COM at mid-length, $I_c = ml^2/12$). (a) Write M(q) and g(q) [5]. (b) The arm is momentarily at rest at q = (45°, −30°) with $\ddot{\mathbf{q}}$ = (1, 2) rad/s^2. Find the joint torques [3].

Q7 · PID gains and stability calculate explain 7 marks

A partitioned joint uses $f' = \ddot{x}_d + k_v\dot{E} + k_pE + k_i\int E\,dt$. (a) With k_p = 49 s^{-2}, choose k_v for critical damping of the PD part [1]. (b) Derive the third-order error equation and the characteristic polynomial [3]. (c) For which k_i is the loop stable? Is k_i = 100 stable? [2] (d) What is the steady-state error under a constant disturbance? [1]

Q8 · Mecanum (Swedish 45°) robot derive calculate 6 marks

Four wheels (r = 0.05 m) at (±0.20, ±0.15) m all drive along x_C (β = 0). Front-left and rear-right rollers have γ = −45°; front-right and rear-left have γ = +45°. (a) Using the generalised model,

write the four wheel rows [3]. (b) Find the wheel speeds for pure sideways motion v = 0.2 m/s [2]. (c) Is the robot holonomic? [1]

Q9 · Planning concepts explain 6 marks

(a) Why are grid planners impractical for a 6-DoF arm? Quantify with 100 cells per joint [2]. (b) Compare PRM and RRT: what they build, when to use each, and their completeness [3]. (c) When is A* guaranteed optimal? [1]

Q10 · Mechanisms: actuation and G–K failure explain 4 marks

(a) Define fully actuated, under-actuated and (kinematically) redundant, with one example each [2]. (b) A planar body is hung from the ground by three equal parallel links. G–K gives M = 0, yet it moves. Explain [2].